THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 108	Find the first multiple of 9.
2	**(a)** 9873	Start from 9876 and test downwards.
3	**(d)** 5	Repeat the routine and count.
4	**(b)** 9	Follow the rule term by term.
5	**(c)** even	Test with small values.
6	**(d)** 21	Triangular numbers: $n(n+1)/2$.
7	**(b)** 3	Add the reverse each time.
8	**(b)** $(10a+b)-(10b+a)$	Difference of a number and its reverse is $9(a-b)$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	2m + 1 + 2n + 1 = 2(m + n + 1), which is even.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	5 + 8 + 3 + 2 = 18 is divisible by 9, so A is true. R is true but does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$11(a+b)=110$ so $a+b=10$; $9(a-b)=36$ so $a-b=4$ (1); $a=7$, $b=3$; Number $=73$ (1).	1+1
12	If n is even, $n(n+1)$ is even $\times$ odd, so even; If n is odd, $n+1$ is even, so the product is even (1); For $n=13$: $13\times14=182$, which is even (1).	1+1
13	Digit sum $=15$; Divisibility by 3 is satisfied by every arrangement (1); Largest arrangement: 7431 (1).	1+1
14	$11(a+b)=143$ so $a+b=13$; $9(a-b)=45$ so $a-b=5$ (1); $a=9$, $b=4$; Number $=94$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Digit sum $=13+2\times*$; Need a multiple of 9 (1); Possible $*$: 7 (1); Numbers: 47376 (1).	1+1+1
16	$5321-1235=4086$; $8640-0468=8172$; $8721-1278=7443$; $7443-3447=3996$; $9963-3699=6264$; $6642-2466=4176$; $7641-1467=6174$ (1); Steps $=7$ and the sequence is $1235, 4086, 8172, 7443, 3996, 6264, 4176, 6174$ (2).	1+2
17	Try 2, 3, 4, ... numbers in turn (1); $7+8$; $4+5+6$; $1+2+3+4+5$ (1); Number of ways $=3$ (1).	1+1+1
	OR: Digit sum $=10+2\times*$; Need a multiple of 9 (1); Possible $*$: 4 (1); Numbers: 74241 (1).	1+1+1
18	$9521-1259=8262$; $8622-2268=6354$; $6543-3456=3087$; $8730-0378=8352$; $8532-2358=6174$ (1); Steps $=5$ and the sequence is $1259, 8262, 6354, 3087, 8352, 6174$ (2).	1+2
	OR: Try 2, 3, 4, ... numbers in turn (1); $10+11$; $6+7+8$; $1+2+3+4+5+6$ (1); Number of ways $=3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $4+5+6$ (1). (ii) $6+7+8=21$ (1). (iii) Yes; the sum is $(n-1)+n+(n+1)=3n$ (1). (iv) $10+11+12=33$ (1) and $16+17=33$ (1).	1+1+1+2
20	$155+551=706$; $706+607=1313$; $1313+3131=4444$ (1); Steps $=3$ (2); Palindrome $=4444$ (2).	1+2+2
	OR: $10a+b=3(a+b)$ (1); Simplify: $7a=2b$ (1); Check digit pairs: numbers 27 (1); Number of such numbers $=1$; their sum $=27$ (2).	1+1+1+2

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a+b$ as $10a+b$ again; the reverse is $10b+a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 7	Repeat the routine and count.
2	**(c)** 17	Follow the rule term by term.
3	**(a)** 3	Test with small values.
4	**(b)** 28	Triangular numbers: $n(n+1)/2$.
5	**(b)** 6	Add the reverse each time.
6	**(c)** $(10a+b)+(10b+a)$	The sum is $11(a+b)$.
7	**(a)** 10008	Find the first multiple of 9.
8	**(c)** 9870	Start from 9876 and test downwards.
9	**(c)** A is true but R is false.	The sum is 11(a + b), so A is true; R is false because the correct result is 11(a + b).
10	**(d)** A is false but R is true.	The product of two odd numbers is odd (e.g. 3 x 5 = 15), so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	If n is even, $n(n+1)$ is even $\times$ odd, so even; If n is odd, $n+1$ is even, so the product is even (1); For $n=24$: $24\times25=600$, which is even (1).	1+1
12	Digit sum $=11$; Divisibility by 5 needs the right last digit (1); Largest arrangement: 6410 (1).	1+1
13	$11(a+b)=121$ so $a+b=11$; $9(a-b)=63$ so $a-b=7$ (1); $a=9$, $b=2$; Number $=92$ (1).	1+1
14	If n is even, $n(n+1)$ is even $\times$ odd, so even; If n is odd, $n+1$ is even, so the product is even (1); For $n=37$: $37\times38=1406$, which is even (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Digit sum $=17+2\times*$; Need a multiple of 9 (1); Possible $*$: 5 (1); Numbers: 55854 (1).	1+1+1
16	$9721-1279=8442$; $8442-2448=5994$; $9954-4599=5355$; $5553-3555=1998$; $9981-1899=8082$; $8820-0288=8532$; $8532-2358=6174$ (1); Steps $=7$ and the sequence is $1279, 8442, 5994, 5355, 1998, 8082, 8532, 6174$ (2).	1+2
17	Try 2, 3, 4, ... numbers in turn (1); $13+14$; $8+9+10$; $2+3+4+5+6+7$ (1); Number of ways $=3$ (1).	1+1+1
	OR: Digit sum $=18+2\times*$; Need a multiple of 9 (1); Possible $*$: $0, 9$ (1); Numbers: $30609, 39699$ (1).	1+1+1
18	$7321-1237=6084$; $8640-0468=8172$; $8721-1278=7443$; $7443-3447=3996$; $9963-3699=6264$; $6642-2466=4176$; $7641-1467=6174$ (1); Steps $=7$ and the sequence is $1327, 6084, 8172, 7443, 3996, 6264, 4176, 6174$ (2).	1+2
	OR: Try 2, 3, 4, ... numbers in turn (1); $16+17$; $10+11+12$; $3+4+5+6+7+8$ (1); Number of ways $=3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $83+38=121$ (1). (ii) $121=11\times11$ (1). (iii) $(10a+b)+(10b+a)=11a+11b=11(a+b)$ (1). (iv) $11\times(6+2)=88$ (1); check: $62+26=88$ (1).	1+1+1+2
20	$174+471=645$; $645+546=1191$; $1191+1911=3102$; $3102+2013=5115$ (1); Steps $=4$ (2); Palindrome $=5115$ (2).	1+2+2
	OR: $10a+b=5(a+b)$ (1); Simplify: $5a=4b$ (1); Check digit pairs: numbers 45 (1); Number of such numbers $=1$; their sum $=45$ (2).	1+1+1+2

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.

- Writing the reverse of $10a + b$ as $10a + b$ again; the reverse is $10b + a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** an odd multiple of 3	Test with small values.
2	**(d)** 36	Triangular numbers: $n(n+1)/2$.
3	**(c)** 4	Add the reverse each time.
4	**(a)** 9876	Start from 9876 and test downwards.
5	**(a)** 1	Repeat the routine and count.
6	**(a)** 20	Follow the rule term by term.
7	**(c)** 11	Test with small values.
8	**(d)** 45	Triangular numbers: $n(n+1)/2$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	24 = 3 x 8.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	6, 3, 10, 5, 16, 8, 4, 2, 1 reaches 1. Both are true, but they are separate facts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Digit sum $= 23$; Divisibility by 2 needs the right last digit (1); Largest arrangement: 9842 (1).	1+1
12	$11(a+b) = 99$ so $a+b=9$; $9(a-b)=27$ so $a-b=3$ (1); $a=6$, $b=3$; Number $= 63$ (1).	1+1
13	If n is even, $n(n+1)$ is even $\times$ odd, so even; If n is odd, $n+1$ is even, so the product is even (1); For $n=50$: $50\times 51 = 2550$, which is even (1).	1+1
14	Digit sum $= 18$; Divisibility by 9 is satisfied by every arrangement (1); Largest arrangement: 8541 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Digit sum $= 14 + 2\times *$; Need a multiple of 9 (1); Possible $*$: 2 (1); Numbers: 82125 (1).	1+1+1
16	$8631-1368=7263$; $7632-2367=5265$; $6552-2556=3996$; $9963-3699=6264$; $6642-2466=4176$; $7641-1467=6174$ (1); Steps $= 6$ and the sequence is $1368, 7263, 5265, 3996, 6264, 4176, 6174$ (2).	1+2
17	Try 2, 3, 4, ... numbers in turn (1); $11+12+13$; $1+2+3+4+5+6+7+8$ (1); Number of ways $= 2$ (1).	1+1+1
	OR: Digit sum $= 12 + 2\times *$; Need a multiple of 9 (1); Possible $*$: 3 (1); Numbers: 23733 (1).	1+1+1
18	$7431-1347=6084$; $8640-0468=8172$; $8721-1278=7443$; $7443-3447=3996$; $9963-3699=6264$; $6642-2466=4176$; $7641-1467=6174$ (1); Steps $= 7$ and the sequence is $1374, 6084, 8172, 7443, 3996, 6264, 4176, 6174$ (2).	1+2
	OR: Try 2, 3, 4, ... numbers in turn (1); $22+23$; $14+15+16$; $7+8+9+10+11$; $5+6+7+8+9+10$; $1+2+3+4+5+6+7+8+9$ (1); Number of ways $= 5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $6+3+8=17$ (1). (ii) $17+*$ must be a multiple of 3, so $*=1, 4$ or 7 (1). (iii) $17+*=18$ gives $*=1$ (1). (iv) The number is 6318 (1); $6318\div 9 = 702$ (1).	1+1+1+2
20	$156+651=807$; $807+708=1515$; $1515+5151=6666$ (1); Steps $= 3$ (2); Palindrome $= 6666$ (2).	1+2+2
	OR: $10a+b = 8(a+b)$ (1); Simplify: $2a = 7b$ (1); Check digit pairs: numbers 72 (1); Number of such numbers $= 1$; their sum $= 72$ (2).	1+1+1+2

Mistakes to avoid:

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- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a+b$ as $10a+b$ again; the reverse is $10b+a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).

- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

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